Dijiste:

UPDATE REPORT MAKE IT SIMPLE EASY FOR INSTRUCTOR TO FOLLOW

Procesó durante 23s >

ChatGPT dijo:

PS6_Genesis_Report_Updated.pdf
PDF

Dijiste:

[MP] Homework 07 of 12 (Machine Problem): Few-Shot Trope/Genre Classification (GenAI)
svg
In Progress
NEXT UP: Submit Assignment
Unlimited Attempts Allowed
Home » Phase 3: Scrub & Embed » Module 06: GenAI & LLMs (Few-Shot Trope/Genre Classification) » MP 7
All Assignments

Critical Submission Policy: Evaluation Source Separation (INV-12) & CPU Baseline (INV-14/15)

Evaluation Source Separation (INV-12): Graded metrics must be computed with `compute_graded_metric` against the instructor-provided evaluation set (`eval_mp7.jsonl`). Metrics on your own splits must use `compute_dev_metric` (`grading_eligible=False`) and are advisory only.
Environment & Quota Constraints (INV-14/INV-15/INV-18): Use an instruction-tuned model small enough for CPU inference or a free-tier hosted API under the team quota killswitch (INV-18); record the exact model identifier and revision hash. All work must be solvable on the pinned CPU DevContainer; GPU access is optional and never required. **Note:** The due date has been extended to **Tue 03 Nov 2026**; all subsequent assignment due dates remain unchanged.

MACHINE PROBLEM 7 · PHASE 3

MP 7: Few-Shot Trope/Genre Classification (GenAI)

Prototype, kNN, and fixed-demonstration LLM classifiers compared under a controlled few-shot protocol

📅 **Assigned:**

Wed 21 Oct 2026

⏰ **Due:**

Fri 30 Oct 2026

Tue 03 Nov 2026, 11:59 PM CST

💯 **Points:**

30

👥 Individual

📂 **Submit:**

Git tag `mp7-submission` + `report.md` (≤3 pp), notebook, `predictions.jsonl`, `GenAI.pdf`

Required Readings

Speech and Language Processing (3rd ed. draft, Jurafsky & Martin, 19 Aug 2026)

- **Chapters 6–8:** Neural networks, large language models, in-context learning, and the transformer architecture.
- **Few-Shot Prompting Mechanics:** Demonstration formatting, label verbalizers, and position bias in autoregressive LMs.

📖 Access Resource →

Hugging Face Transformers Documentation

- **"Text Classification" Task Guide:** Tokenization, sequence-classification heads, the `Trainer` workflow, constrained decoding, and metric computation.

📘 Access Resource →

1. Overview & Pipeline Lineage

Build an embedding index from provenance-admitted chunks (FAISS `IndexFlatIP`) and compare three classifiers on the instructor-provided evaluation set: **(i) embedding-prototype**, **(ii) kNN over the exemplar index**, and **(iii) few-shot LLM prompting with fixed, class-balanced demonstrations**. Retrieval-conditioned demonstration selection is deferred to MP 9; calibration, significance testing, and error taxonomy are deferred to MP 10. Report **macro-F1 with 95% bootstrap CIs only**.

- **VDB Track Label Space:** 8–12 trope/genre labels from the ontology (**CIS 531:** instructor-provided unmodified schema per INV-11; **CIS 731:** your custom PS 6 ontology).
- **ASR Track Label Space:** Drawn from the instructor-provided `asr_label_set.yaml`.
- **Upstream Inputs:** PS 6 schema and label space; MP 5 chunk records; MP 3 corpus.
- **Downstream Consumers:** MP 9 (reuses the `Classifier` interface, prompt harness, and exemplar index; the fixed-demonstration result is the baseline for retrieval-augmented comparison); MP 10 (best model is interpreted); PS 8 (preliminary baselines).

🤖 Initiating Your Draft: The 5-Stage Generative AI Kit Procedure & MP 7 Tollbooth Prompt (Click to Expand)
Use the **Generative AI Kit (v3.0)** to help structure your submission. Follow these five stages:

1. **Protocol Lock:** Commit `protocol.md` (label space, shot sweep $k \in \{0, 1, 3, 5, 8\}$, at least 5 seeds, bootstrap CI method) *before* generating any results.
2. **Prompt Template Synthesis:** Draft the instruction, demonstration format, label verbalizers, constrained decoding, and output parser with the AI Kit; audit for label leakage and demonstration-position bias.
3. **Index and Baseline Build:** Embed admitted chunks, build the FAISS `IndexFlatIP` index, and implement the prototype and kNN baselines.

4. **Sweep Execution:** Run the shot sweep across seeds with quota logging; compute graded metrics only via `compute_graded_metric`.
5. **Results Audit:** Re-derive each reported number from `predictions.jsonl`; reject any AI-generated analysis that cannot be reproduced.

[MP 7 TOLLBOOTH PROMPT] I am working on MP 7. I need to build the prompt template for few-shot in-context learning to classify my vector chunks. *Here is the Toll:* [Provide your specific 8-12 trope/genre label space or asr_label_set.yaml labels]. *Properties:* My chosen shot sweep k values are {0, 1, 3, 5, 8} across >=5 seeds, and my instruction-tuned model identifier is [e.g., Qwen/Qwen2.5-1.5B-Instruct or meta-llama/Llama-3.2-3B-Instruct]. Do not write the Hugging Face pipeline code. Help me design the strict formatting constraints, verbalizers, and system instructions for the LLM prompt to ensure the output perfectly matches the label space without hallucinating labels. Suggest concrete controls for majority-label bias, demonstration-position bias (including permutation), and label leakage.

Part 1: Experimental Design & Prompt Failure-Mode Controls

(30 Points)
Before running your shot sweep, lock your experimental protocol and engineer your prompt harness against known in-context learning failure modes. Document both sections in `report.md` (≤3 pages total).

A. Protocol and Label Space (15 Points)

- Commit `protocol.md` prior to generating experimental predictions and include its exact specification in `report.md`.
- Explicitly define your 8–12 label space (VDB track: instructor ontology for CIS 531 or PS 6 custom ontology for CIS 731; ASR track: `asr_label_set.yaml`), your shot sweep $k \in \{0, 1, 3, 5, 8\}$, your random seed list (≥5 seeds), and your 95% bootstrap confidence interval procedure.

B. Prompt Specification and Failure-Mode Controls (15 Points)

- Provide the exact prompt template, demonstration format, label verbalizers, constrained decoding rules, and output parser.
- Analyze three specific failure modes—**(1) majority-label bias**, **(2) demonstration-position bias**, and **(3) label leakage**—and implement a concrete, verifiable control for each, explicitly including **demonstration-order permutation** across seeds/runs.

Part 2: Classifier Implementation, Shot-Sweep Rigor & Reproducibility

(70 Points: 65 Points Technical Implementation + 5 Points GenAI Audit)
Objective: Implement a unified 3-classifier benchmark over your FAISS `IndexFlatIP` index, enforce runtime contracts via `deal`, and evaluate macro-F1 across the shot sweep on the instructor-provided evaluation set.

1. **Classifier Implementations & Contracts (25 pts):** Implement all three classifiers—(i) embedding-prototype, (ii) kNN over the FAISS `IndexFlatIP` exemplar index, and (iii) fixed-demonstration few-shot LLM—behind a single unified `Classifier` interface. Enforce `deal` post-conditions on `predict_proba`: all probabilities are non-negative, row sums equal `1.0` within `1e-6`, and all predicted labels belong strictly to the declared label space.
2. **Shot-Sweep Execution & Statistical Reporting (25 pts):** Execute the sweep for $k \in \{0, 1, 3, 5, 8\}$ across at least 5 random seeds. Compute graded metrics exclusively via `compute_graded_metric` on the instructor evaluation set. Report **macro-F1 (mean ± std over seeds and 95% bootstrap CI)** per classifier and shot count, plus per-class precision and recall at the best configuration.
3. **Reproducibility & Quota Log (15 pts):** Record the pinned DevContainer environment, exact Hugging Face / API model identifier and commit/revision hash, total API token/request

usage versus quota (INV-18), and verification that the entire pipeline runs cleanly on the CPU baseline (INV-14/INV-15).

4. **GenAI Audit Verification (5 pts):** Include your `GenAI.pdf` log documenting all prompts, tollbooth inputs, and raw outputs, or explicitly state `"No GenAI Used."`

Deliverable: Push Git tag `mp7-submission` containing `protocol.md`, `report.md` (at most 3 pages), your executed Jupyter notebook (`.ipynb`), `predictions.jsonl`, and `GenAI.pdf`, and submit your repository tag URL and PDF report on Canvas.

Scoring Rubric

Component	Raw Weight	Norm (30)	Description
Protocol and Label Space	15 pts	4.5 pts	`protocol.md` is complete (8–12 ontology/ASR labels, $k \in \{0,1,3,5,8\}$, ≥5 seeds, bootstrap CI method) and committed prior to results generation.
Prompt Specification & Failure Controls	15 pts	4.5 pts	Prompt template, verbalizers, and parser are precise; concrete controls implemented for majority-label bias, demonstration-position bias (with permutation), and label leakage.

Component	Raw Weight	Norm (30)	Description
Classifier Implementation & Contracts	25 pts	7.5 pts	Prototype, kNN (FAISS `IndexFlatIP`), and few-shot LLM implemented behind one `Classifier` interface; `deal` contracts on `predict_proba` hold on all outputs.
Shot-Sweep Rigor & Source Separation	25 pts	7.5 pts	Macro-F1 (mean ± std over ≥5 seeds and 95% bootstrap CIs) and best-config per-class P/R computed strictly via `compute_graded_metric` (INV-12) and reproducible from `predictions.jsonl`.
Reproducibility & Quota Log	15 pts	4.5 pts	Pinned environment (INV-14), exact model ID and revision hash, API quota log (INV-18), and verified CPU baseline execution (INV-15).
GenAI Verification	5 pts	1.5 pts	Includes complete `GenAI.pdf` audit log or explicit attestation that no Generative AI was used.

🔭 Downstream Assignment Specifications (PS 8, MP 9, MP 10, PS 11, HW 12 — Unchanged Due Dates)

- **PS 8: Interim Report (Mandatory Milestone)** — *Released:* Thu 29 Oct 2026 | *Due:* **Sat 07 Nov 2026** (30 pts, Team/Solo). Single PDF (≤10 pp) + figure sources in `interim/` under tag `ps8-submission`: (1) Team & Taxonomy Metadata + Integrity

Attestation (2 pts), (2) Scope Delta & Proposal Responses (4 pts), (3) Data Provenance & Exploratory Baselines with ≥3 figures including UMAP sensitivity sweep over `n_neighbors` /`min_dist` and MP 7 baseline (10 pts), (4) Architecture, Invariants & `deal` Excerpts (5 pts), (5) Quantified Risk Register (≥5 risks) & 50–200 Annotation Plan (5 pts), (6) GitHub Repo Link (2 pts), (7) `GenAI.pdf` (2 pts).

- **MP 9: Retrieval-Augmented Classification & Convolutions** — *Released:* Sat 07 Nov 2026 | *Due:* **Mon 16 Nov 2026** (100 pts / 30 norm, Individual with partner overlap). Strict CI ceiling: **at most 50 gold objects or 2 hours** (`gold_raw.*` > 50 rows fails CI with 0). Report Cohen's κ + 95% CI on 20-object partner overlap (`overlap_ids.txt`), weak-supervision expansion (≥1,000 chunks marked `source=silver`), top-*k* exemplar RAG classification ($k \in \{3, 5, 10\}$) vs. MP 7 baseline, and cohort convolution (`[0.25, 0.5, 0.25]` smoothing and `[-1, 1]` first-difference kernels; **CIS 731:** learned 1-D PyTorch conv kernel) under tag `mp9-submission` .
- **MP 10: Data Science Interpretability & Metrics Scoring** — *Released:* Mon 16 Nov 2026 | *Due:* **Wed 25 Nov 2026** (100 pts / 30 norm, Individual). Build `harness/` with `evaluate(model, X, y)` and `deal` contracts on adjudicated MP 9 gold set; derive micro/macro/weighted F1 (with imbalance counterexample), P@k, Recall@k, MRR, nDCG@k, and NIST OpenASR WER; global permutation importance (≥30 repeats) + local retrieval explanations for ≥10 misclassified examples; ECE under 10 and 15 bins + post-hoc calibration; paired bootstrap ($B \geq 10{,}000$) or McNemar's test with Benjamini–Hochberg FDR control across ≥6 comparisons; CI-width vs. *n* on silver subsamples under tag `mp10-submission` .
- **PS 11: Ethics, Data Storytelling, & Dashboards** — *Released:* Wed 25 Nov 2026 | *Due:* **Fri 04 Dec 2026** (100 pts / 30 norm, Individual). Ethics memo PDF (≤4 pp) + 5-view Wilke storyboard + restart-and-run-all prototype dashboard + limitations panel under tag `ps11-submission` .
- **HW 12: Final Project Deliverables (Mandatory Milestone)** — *Released:* Sat 05 Dec 2026 | *Due:* **Mon 14 Dec 2026** (200 pts / 300 Substance + 30 Milestone, Team/Solo). `REPORT.pdf` (≤12

pp) + digest-pinned MLflow reproducibility package (`make verify` , 50–200 gold range gate, serving path + PSI/KS drift runbook) under tag `hw12-final` .

🏠 Return to Course Home

in order to complete this assigment, make me a break down like you did for ps5

ChatGPT dijo:

Dijiste:

code literacy needed to suceed

ChatGPT dijo:

Dijiste:

```
{
  "youth": "youth and growing up",
  "friendship": "friendship and loyalty",
  "romance": "love and romance",
  "self_love": "self-love and identity",
  "longing": "loss and longing",
  "fame": "fame and success",
  "social": "social commentary",
  "celebration": "celebration and fun"
}
```

this is my labels.json on data, with this said could you provide the make_data.py, eval_mp7.jsonl and examplars.jsonl

ChatGPT dijo:

Dijiste:

he did not provide eval_mp7.jsonl, so i creating my own

ChatGPT dijo:

Dijiste:

how i will run it to make sure everything is ok?

ChatGPT dijo:

Dijiste:

```
genesisjoseph@Laptop-de-Genesis MP7_Genesis % python3
make_data.py
96 exemplars, 64 eval, 8 labels
genesisjoseph@Laptop-de-Genesis MP7_Genesis %
```

ChatGPT dijo:

Dijiste:

```
genesisjoseph@Laptop-de-Genesis MP7_Genesis % wc -l
data/*.jsonl
      64 data/eval_mp7.jsonl
      96 data/exemplars.jsonl
     160 total
genesisjoseph@Laptop-de-Genesis MP7_Genesis % python3 -m
pytest -q
```

ChatGPT dijo:

Dijiste:

MP7_Genesis.zip
Archivo

HERE IS MY MP7. base solely in what i have, create the report and protocol allowed for this course

Procesó durante 45s >

ChatGPT dijo:

MP7_Report.pdf
PDF

report.md
Documento

protocol.md
Documento